Chiral Phase-Shear Binding (CPSB): Rendering Syntax as Gauge-Theoretic Tension in Wave-Based SI

Irfan Mahir — Independent Researcher, Dhaka, Bangladesh — irfan@furylogic.com

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Irfan Mahir *Independent Researcher, Dhaka, Bangladesh* irfan@furylogic.com

Abstract In wave-based Synthetic Intelligence, continuous phase-sweep emission (CPME) frees output length from hardcoded templates but degrades syntax into emergent stochastic ordering. To elevate language construction from a "loose variable" to a physically guaranteed structure, we introduce **Chiral Phase-Shear Binding (CPSB)**. By modifying the Kuramoto coupling to act as a Gauge Field, relational tunnels actively enforce directed phase offsets between nodes. Syntax emerges not from fixed emission slots, but from the rigid topological tension (Ψ) embedded within the graph's edges, physically forcing subjects to lead verbs, and verbs to lead objects during carrier-wave projection.

1. Introduction: The Binding Problem in SI

Currently, verb-material cavities float to their VGHT band-top, and output order depends entirely on their emergent atan2 phase. There is no physical law forcing `notch` to precede `minecraft`. In a true biological substrate, the relation itself ("created") exerts a directional shear force on the concepts it connects. We must shift from scalar coupling (which just pulls phases together) to **Gauge-Coupling**, which pulls phases into a specific grammatical alignment.

2. Novel Architecture: CPSB Mechanisms

2.1. Gauge-Theoretic Kuramoto Offset (Ψ_{ij})

Standard Kuramoto tries to perfectly synchronize two connected cavities: $\sin(\theta_j - \theta_i)$. We introduce a syntactic phase-offset Ψ_{ij} assigned to specific tunnel types (the axioms). The new

phase-update equation becomes:

$$\frac{d\theta_i}{dt} = \omega_i + \sum K_{ij} \sin(\theta_j - \theta_i - \Psi_{ij})$$

- **Mechanism:** If `notch` (i) and `minecraft` (j) are connected by a `REL_CAUSES` tunnel, that tunnel has an intrinsic $\Psi_{ij} = -\frac{\pi}{3}$.
- **Result:** The physics engine will no longer try to make their phases equal. It will physically force `notch` to sit exactly $\frac{\pi}{3}$ radians ahead of `minecraft`.

2.2. The Lexical Hinge (Verb-Pinning)

A verb is not just a loose cavity floating in a band; it is the physical hinge between the Subject and the Object.

- **Mechanism:** When a cross-cluster tunnel opens (VGHT), the tunnel structurally binds the verb cavity (e.g., `created`) to the source and target.
- The Subject is pinned to $\theta_{\text{verb}} + \Delta\phi$.
- The Object is pinned to $\theta_{\text{verb}} - \Delta\phi$.
- **Result:** As the CPME carrier wave sweeps from 0 to 2π, it will *inevitably* hit the Subject first, the Verb second, and the Object third. The sentence structure is mathematically guaranteed by the phase tension, without ever defining a "Subject Slot".

2.3. Phase-Consistent Hebbian Consolidation (The CPME-II Hook)

To ensure the SI doesn't just read syntax natively but *learns* it natively from user input, we execute the CPME-II Hebbian update. Instead of updating tunnel weight Δw based purely on amplitude, the substrate writes the observed phase difference directly into the graph:

$$\Delta w_{ij} \propto v_i \cdot v_j \cdot \cos(\theta_i - \theta_j - \Psi_{\text{target}})$$

- **Result:** When you type `insulin lowers glucose`, the injection chirality ensures `insulin` arrives first. The Hebbian rule permanently wires that phase-lead into the connection. The SI literally remembers the "direction" of the concept.

3. Expected Dynamics

1. **Input:** `metformin treats diabetes`
2. **Resonance:** The nodes awaken. The `treats` tunnel applies the CPSB gauge force. `metformin` is pushed to $\phi = 1.0$, `treats` is anchored at $\phi = 2.0$, and `diabetes` is pushed to $\phi = 3.0$.
3. **Emission:** The CPME sweep passes over them. They emit in perfect S-V-O order.
4. **No Templates:** If the input is just `glucose` causing a cascade to `insulin`, the phase offsets still apply downstream, resulting in a syntactically correct multi-token chain.
